

NIRAV SURABHI

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EDUCATION

Purdue University

Bachelor of Science, Mechanical Engineering

West Lafayette, IN
Expected Graduation: May 2029

Independence High School

High School Diploma, GPA: 3.8/4.00

Loudoun County, VA
June 2026

Relevant Coursework: AP Physics C (Mechanics and E&M), AP Computer Science A, Multivariable Calculus

Honors: Mathematics Scholar, Central Virginia Governor's School for Science & Technology; Presidential Volunteer Service Award, Silver; 1st Place, Aguado Guitar Competition; All-Virginia Guitar Ensemble

PROFESSIONAL EXPERIENCE

I3 Simulations — CAD Intern

Jun 2025 – May 2026

- Designed 3D CAD models and prototyped haptic-feedback systems with cross-functional developers for immersive medical-training simulators.
- Conducted prototype verification testing for usability, reliability, and conformance to design requirements.

The Bike Lane — Bike Mechanic (Intern, promoted to Part-Time Staff)

Jun 2021 – Aug 2025

- Assembled bicycles from the frame up and ran the final safety inspection on every build before customer delivery.
- Diagnosed root causes on nonconforming units and repaired drivetrain, brake, and shifting systems.

Ember Learning (formerly DeAP Learning Labs) — AI Engineer & Software Intern

Feb 2021 – 2022

- Engineered the LLM prompts behind AP-tutor chatbots that replicate the teaching styles of well-known YouTube AP educators; the team secured a Microsoft and OpenAI grant and reached a \$1M+ valuation.
- Built an automated prompt-testing suite that graded AP DBQ and FRQ essays against the official rubrics, scoring both answer accuracy and reasoning.
- Indexed lecture-video transcripts in a vector database so each student question retrieves the exact video and timestamp.

RESEARCH AND PROJECTS

Passive Vortex Propulsion in Air: A Proof of Concept Demonstration — IEEE EngiTek 2025 — First Author 2024 – Dec 2025

- Built a wind tunnel and test rig with a soldered, calibrated Arduino load-cell system; measured 3.1 N mean net thrust over 20 trials ($p < 0.001$), the first experimental evidence of passive vortex propulsion in air.
- Compared flat and NACA 0024/0012/0009 fins using ANOVA and Dunn's test in Python plus CFD in Fusion 360; won the AIAA Aerospace Award at the Regional Science & Engineering Fair (invitation to NASA Goddard).

SpareMe — spareme.com — Founder & Engineer

Jul 2026 – Present

- Shipped an iPhone app to the App Store that delegates phone calls to an AI agent: it dials U.S. businesses, works phone trees, waits on hold, and returns a written outcome; users watch a live transcript and can type instructions, listen in, or take over mid-call.
- Built the SwiftUI client and Node/TypeScript backend on Twilio ConversationRelay (real-time speech) and OpenAI (reasoning), with verified caller ID, encrypted PostgreSQL case state, APNs, StoreKit per-minute billing with signed transaction verification, and fail-closed kill switches and rate limits.

Oh — github.com/21J3phy/Oh — Creator, open source

May 2026 – Jun 2026

- Built shared memory for AI coding agents: hook-driven capture of Claude Code and Codex sessions, local secret scrubbing, pgvector retrieval, and an MCP tool that answers why code is the way it is without interrupting its author.

Motor-Based Resistance Trainer — Designer & Builder

Mar 2025

- Built three prototypes (eddy-current, chain-driven, geared) of a portable, battery-powered strength trainer; a 188:1 geared motor under Arduino PWM control gave a linear, repeatable force curve ($R^2 = 0.993$) for \$215.

LEADERSHIP AND INVOLVEMENT

FIRST Tech Challenge Robotics, Absolute Zero #12096 — Team Captain & Lead Programmer

Sep 2022 – Jun 2026

- Led the team for four seasons; designed and built every mechanism in Fusion 360 and advanced to the Chesapeake regional championship.
- Wrote the autonomous and tele-op code in Java, integrating PID control loops, sensor fusion (encoders, gyroscopes), and odometry.

Purdue Formula SAE — Team Member

Aug 2026 – Present

- Joined the vehicle team in Fall 2026 and completed the design-onboarding sprint in CAD.

Science Olympiad, Independence High School — Team Captain

- Engineered a PID-controlled, 3D-printed Arduino electric vehicle for timed distance runs and an electrode sensor that measures water salinity.

RELEVANT SKILLS

Software: Python, Java, C++, Swift, TypeScript, SQL; SwiftUI, Next.js, Node.js, FastAPI, PostgreSQL, Twilio, StoreKit, APNs, Git, CI/CD

AI / ML: PyTorch, computer vision, LLM agents and tool use, vector search (pgvector), prompt design, evaluation

Hardware: Fusion 360 (CAD, FEA, CFD), 3D printing, Arduino, load cells, soldering, PID / PWM control, sensor fusion